Clustering, $m = 5$ labelled windows per cluster

Table 1: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.45	**0.36**	0.67	**0.34**	0.04	0.01	0.38	0.24
k-means	Sensor+ToD hist.	**0.56**	0.31	**0.68**	0.23	**0.37**	**0.20**	**0.54**	**0.24**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.70**	**0.47**	**0.69**	0.34	0.09	0.06	0.49	0.29
k-means	Sensor+ToD hist.	0.51	0.32	**0.69**	**0.37**	**0.43**	**0.27**	**0.54**	**0.32**

Table 2: Clustering, L1, all classes; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.45	**0.27**	0.66	**0.24**	0.04	0.01	0.38	0.17
k-means	Sensor+ToD hist.	**0.56**	0.23	**0.68**	0.16	**0.36**	**0.18**	**0.53**	**0.19**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.69**	**0.35**	**0.69**	0.24	0.09	0.05	0.49	0.21
k-means	Sensor+ToD hist.	0.50	0.24	**0.69**	**0.26**	**0.43**	**0.25**	**0.54**	**0.25**

Table 3: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	0.48	**0.29**	0.66	**0.30**	0.06	0.01	0.40	**0.20**
k-means	Sensor+ToD hist.	**0.56**	0.25	**0.68**	0.20	**0.41**	**0.15**	**0.55**	**0.20**
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.69**	**0.33**	**0.69**	0.30	0.14	0.05	0.51	0.23
k-means	Sensor+ToD hist.	0.54	0.27	**0.69**	**0.32**	**0.48**	**0.25**	**0.57**	**0.28**

Clustering, $m = 1$ labelled windows per cluster

Table 4: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.53**	**0.37**	**0.78**	**0.47**	0.11	0.07	**0.47**	**0.30**
k-means	Sensor+ToD hist.	0.43	0.27	0.45	0.15	**0.43**	**0.25**	0.44	0.22
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.65**	**0.37**	**0.69**	**0.34**	0.13	0.08	**0.49**	0.26
k-means	Sensor+ToD hist.	0.55	0.36	0.45	0.30	**0.34**	**0.23**	0.45	**0.30**

Table 5: Clustering, L1, all classes; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.53**	**0.27**	**0.78**	**0.33**	0.11	0.06	**0.47**	**0.22**
k-means	Sensor+ToD hist.	0.42	0.20	0.45	0.10	**0.42**	**0.22**	0.43	0.17
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.65**	**0.33**	**0.68**	**0.24**	0.13	0.07	**0.49**	0.21
k-means	Sensor+ToD hist.	0.55	0.27	0.45	0.21	**0.34**	**0.21**	0.45	**0.23**

Table 6: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.54**	**0.27**	**0.78**	**0.41**	0.36	0.12	**0.56**	**0.27**
k-means	Sensor+ToD hist.	0.44	0.21	0.45	0.13	**0.48**	**0.22**	0.46	0.19
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.65**	**0.29**	**0.68**	**0.30**	0.35	0.11	**0.56**	**0.23**
k-means	Sensor+ToD hist.	0.55	0.26	0.46	0.26	**0.38**	**0.17**	0.46	**0.23**

Clustering, $m = all$ labelled windows per cluster

Table 7: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.75**	**0.49**	**0.78**	**0.38**	0.00	0.00	**0.51**	**0.29**
k-means	Sensor+ToD hist.	0.72	0.39	0.72	0.24	**0.08**	**0.07**	**0.51**	0.23
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.76**	**0.48**	**0.81**	**0.55**	0.02	0.02	0.53	**0.35**
k-means	Sensor+ToD hist.	0.76	0.47	0.76	0.35	**0.15**	**0.11**	**0.56**	0.31

Table 8: Clustering, L1, all classes; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.75**	**0.36**	**0.77**	**0.26**	0.00	0.00	**0.51**	**0.21**
k-means	Sensor+ToD hist.	0.72	0.29	0.72	0.17	**0.08**	**0.06**	**0.51**	0.17
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.76**	**0.41**	**0.81**	**0.39**	0.02	0.02	0.53	**0.27**
k-means	Sensor+ToD hist.	0.75	0.35	0.76	0.24	**0.15**	**0.10**	**0.56**	0.23

Table 9: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.75**	**0.34**	**0.77**	**0.33**	0.00	0.00	**0.51**	**0.22**
k-means	Sensor+ToD hist.	0.72	0.30	0.72	0.21	**0.09**	**0.08**	**0.51**	0.20
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.77**	0.34	**0.82**	**0.44**	0.03	0.02	0.54	**0.27**
k-means	Sensor+ToD hist.	0.76	**0.35**	0.76	0.31	**0.16**	**0.14**	**0.56**	0.26

Linear probes (logistic regression on frozen representations)

Table 10: Linear probes, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.65**	0.34	**0.78**	**0.40**	0.07	0.05	0.50	0.26
Linear probe	Sensor+ToD hist.	0.64	**0.35**	0.77	0.39	**0.24**	**0.14**	**0.55**	**0.29**
250 random labelled windows									
Linear probe	Sensor hist.	**0.71**	**0.45**	**0.78**	**0.45**	0.06	0.09	0.52	0.33
Linear probe	Sensor+ToD hist.	0.69	**0.45**	0.77	0.43	**0.35**	**0.30**	**0.60**	**0.40**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.78	0.55	**0.83**	**0.56**	0.02	0.02	0.54	0.38
Linear probe	Sensor+ToD hist.	**0.79**	**0.58**	**0.83**	**0.56**	**0.30**	**0.26**	**0.64**	**0.47**

Table 11: Linear probes, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.64**	0.26	**0.78**	**0.28**	0.07	0.06	0.50	0.20
Linear probe	Sensor+ToD hist.	0.63	**0.27**	0.77	0.28	**0.24**	**0.14**	**0.55**	**0.23**
250 random labelled windows									
Linear probe	Sensor hist.	**0.70**	**0.35**	**0.78**	**0.32**	0.06	0.08	0.51	0.25
Linear probe	Sensor+ToD hist.	0.69	**0.36**	0.77	0.32	**0.35**	**0.27**	**0.60**	**0.32**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.77	**0.46**	**0.83**	**0.39**	0.02	0.02	0.54	0.29
Linear probe	Sensor+ToD hist.	**0.79**	0.45	**0.83**	**0.39**	**0.30**	**0.23**	**0.64**	**0.36**

Table 12: Linear probes, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.67**	**0.28**	**0.78**	**0.34**	0.24	0.10	0.56	0.24
Linear probe	Sensor+ToD hist.	0.65	**0.28**	0.77	0.34	**0.31**	**0.13**	**0.58**	**0.25**
250 random labelled windows									
Linear probe	Sensor hist.	**0.71**	0.35	**0.78**	**0.38**	0.20	0.09	0.56	0.27
Linear probe	Sensor+ToD hist.	0.70	**0.35**	0.77	**0.37**	**0.39**	**0.23**	**0.62**	**0.32**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.78	0.41	**0.83**	**0.44**	0.08	0.04	0.57	0.30
Linear probe	Sensor+ToD hist.	**0.80**	**0.47**	0.83	0.43	**0.35**	**0.20**	**0.66**	**0.37**

Linear probe vs clustering at matched label budgets

Table 13: Probe vs clustering at matched label budgets, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.38	0.24	0.50	0.26	0.49	0.29	0.52	0.33	0.53	**0.35**	0.54	0.38
k-means / probe	Sensor+ToD hist.	**0.54**	**0.24**	**0.55**	**0.29**	**0.54**	**0.32**	**0.60**	**0.40**	**0.56**	0.31	**0.64**	**0.47**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–

Table 14: Probe vs clustering at matched label budgets, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.38	0.17	0.50	0.20	0.49	0.21	0.51	0.25	0.53	**0.27**	0.54	0.29
k-means / probe	Sensor+ToD hist.	**0.53**	**0.19**	**0.55**	**0.23**	**0.54**	**0.25**	**0.60**	**0.32**	**0.56**	0.23	**0.64**	**0.36**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–

Table 15: Probe vs clustering at matched label budgets, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.40	**0.20**	0.56	0.24	0.51	0.23	0.56	0.27	0.54	**0.27**	0.57	0.30
k-means / probe	Sensor+ToD hist.	**0.55**	**0.20**	**0.58**	**0.25**	**0.57**	**0.28**	**0.62**	**0.32**	**0.56**	0.26	**0.66**	**0.37**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–